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on

“AGENTIC & GENERATIVE ARTIFICIAL INTELLIGENCE”

Project Title:

“AI SKIN SPECIALIST: VOICE & VISION BASED MULTIMODAL MEDICAL CONSULTATION CHATBOT”

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CERTIFICATE

This is to certify that Ms. Manpreet Kaur, student of Bachelor of Technology in Computer Science & Engineering (B.Tech CSE), has successfully completed her 4-Week Institutional / Industrial Training Programme (TR-102) at DigiMantra Labs as a part of her academic curriculum during June–July, 2026.

During the training period, she worked on the project titled "AI Skin Specialist: Voice & Vision Based Multimodal Medical Consultation Chatbot" and gained practical exposure to relevant technologies, tools, and concepts. The training provided an opportunity to apply theoretical computer science knowledge to real-world problems and develop professional and technical skills.

The successful completion of this training reflects her sincere efforts, dedication, and commitment towards learning and engineering project development.

The work documented in this report represents the training work undertaken by the candidate and is submitted for academic evaluation.

The project involved the design and development of an end-to-end multimodal artificial intelligence system for preliminary dermatological consultation assistance. The implemented application combines speech-to-text recognition via OpenAI Whisper, visual clinical analysis using multimodal Large Language Models (Qwen Vision on Groq), and neural text-to-speech voice generation via Deepgram Aura inside a responsive Gradio web interface.

The student gratefully acknowledges the expert guidance and continuous support of the Head Coordinator, Dr. Kuljit Kaur, throughout the training and project evaluation.

Date: _____

Place: _____

Training Mentor / Supervisor **Head Coordinator**

CANDIDATE'S DECLARATION

I hereby declare that the training report entitled “AI Skin Specialist: Voice & Vision Based Multimodal Medical Consultation Chatbot” submitted for the 4-week training (TR-102) is an authentic record of the work carried out by me during the training period under the guidance and supervision of the concerned mentors.

The project implementation, system design, prompt structures, technical details, and discussions presented in this report are based on the work undertaken and the source files developed and submitted as part of the training project.

I further declare that this report has been prepared solely for academic purposes and represents my own work and understanding. All external technical references, resources, and supporting materials consulted during the development and documentation of the project have been duly acknowledged in the References section.

I take full responsibility for the accuracy and authenticity of the information presented in this report.

Signature of Candidate: _____

Name: Manpreet Kaur
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ABSTRACT

Access to immediate dermatological guidance is frequently hindered by long clinical waiting times, geographic barriers, and shortages of certified specialists in primary care. Patients facing common skin issues—such as rashes, acne flares, or localized lesions—often turn to unverified internet searches, resulting in incorrect self-medication and heightened anxiety. Traditional healthcare software relies on static questionnaires or standalone image classifiers that cannot hold a natural clinical conversation or provide spoken reassurance.

To resolve this challenge, the project "AI Skin Specialist: Voice & Vision Based Multimodal Medical Consultation Chatbot" was developed as a unified, voice-driven consultation assistant. The application allows patients to verbally articulate their symptoms through a microphone or audio upload, accompanied by a photograph or short mobile video clip of the affected skin area. The system orchestrates an end-to-end multimodal pipeline integrating Automatic Speech Recognition (ASR), Multimodal Large Language Models (MLLMs), and Neural Text-to-Speech (TTS) synthesis within an interactive web portal.

The implementation is built on Python 3.13 and leverages low-latency cloud inference via Groq Cloud and Deepgram. Patient audio is transcribed into text using OpenAI Whisper (whisper-large-v3). Visual inputs are processed in-memory using Pillow (PIL) and OpenCV (cv2) for automated video frame extraction and Base64 serialization. The transcribed clinical text and visual data are evaluated by Groq's hosted Qwen Vision model, governed by carefully engineered clinical prompts that enforce concise 2–3 sentence recommendations without markdown artifacts. Internal reasoning deliberation tokens (<think> tags) are stripped via regular expression post-processing before guidance is synthesized into realistic spoken doctor audio via Deepgram Aura (aura-asteria-en).

The web application is implemented in Gradio Blocks with custom medical-grade CSS and prominent safety disclaimers. End-to-end testing confirms sub-3-second inference turnaround, validating the potential of multimodal AI to deliver accessible, conversational first-line healthcare guidance.

ACKNOWLEDGMENT

I would like to express my sincere gratitude to everyone who contributed to the successful completion of my 4-week institutional training programme (TR-102) and the development of this project.

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I am also deeply thankful to DigiMantra Labs and its mentors for providing the technical resources, modern development environment, and practical industry exposure necessary to complete this training. The training provided comprehensive hands-on experience in Artificial Intelligence, Multimodal Deep Learning, Speech Systems, Computer Vision, and full-stack web integration.

I acknowledge the official technical documentation and developer communities of Python, Groq, Deepgram, OpenCV, and Gradio, which served as vital reference points throughout implementation.

Finally, I extend my heartfelt gratitude to my family and friends for their continuous moral support, understanding, and encouragement throughout this training period.

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NOTATIONS / NOMENCLATURE

Term / Acronym	Definition / Meaning
AI	Artificial Intelligence
ML	Machine Learning
LLM	Large Language Model
VLM	Vision-Language Model
MLLM	Multimodal Large Language Model
ASR	Automatic Speech Recognition
STT	Speech-to-Text
TTS	Text-to-Speech
API	Application Programming Interface
REST	Representational State Transfer
LPU	Language Processing Unit (Groq Hardware Architecture)
PIL	Python Imaging Library (Pillow)
OpenCV	Open Source Computer Vision Library
FPS	Frames Per Second
RGB	Red, Green, Blue Color Space
BGR	Blue, Green, Red Color Space
URI	Uniform Resource Identifier
gTTS	Google Text-to-Speech
UI / UX	User Interface / User Experience
SDK	Software Development Kit
CORS	Cross-Origin Resource Sharing

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CHAPTER 1
INTRODUCTION

1.1 Introduction

The integration of Artificial Intelligence (AI) in clinical diagnostics and healthcare consultation represents one of the most transformative advances in modern computer science. Historically, medical consultation and telemedicine systems have relied heavily upon structured questionnaires, manual data entry, and scheduled telephonic consultations. While effective, these conventional approaches frequently suffer from severe bottlenecks, including lengthy scheduling queues, high operational overhead, and substantial geographic barriers that restrict patient access to certified specialists.

Dermatology is fundamentally an audio-visual medical discipline. Patients naturally articulate their concerns through spoken descriptions—detailing the sensation, duration, itching, pain, and progression of a skin lesion—while simultaneously presenting visual manifestations through photographs or video clips. Standard text-based chatbots fail to capture the nuance of patient speech, while standalone computer vision classification models lack the conversational ability to reassure the patient, provide contextual guidance, and deliver spoken advice.

This four-week industrial and institutional training project (TR-102) addresses these limitations by developing the AI Skin Specialist, an intelligent, multimodal voice-driven healthcare consultation assistant. The system provides an end-to-end automated workflow: the patient speaks their symptoms into a microphone or uploads an audio clip, uploads an image or video of the affected skin, and receives both a real-time transcript, clinical text guidance, and an empathetic doctor voice audio response generated in real time.

1.2 Current Scenario in Healthcare & AI

The current landscape of digital healthcare is characterized by an explosion of generative artificial intelligence and multimodal language models. State-of-the-art foundation models are no longer restricted to pure textual processing; they now exhibit cross-modal reasoning

capabilities, allowing them to jointly analyze high-resolution images, video frames, and natural language prompts.

Simultaneously, advancements in automatic speech recognition (ASR) have been revolutionized by deep transformer-based architectures such as OpenAI's Whisper, which demonstrate human-level speech transcription accuracy across diverse accents, ambient noise levels, and specialized medical terminologies. In parallel, neural Text-to-Speech (TTS) models, such as Deepgram Aura, have superseded robotic concatenative synthesizers by producing natural human prosody, emotional inflections, and conversational cadence with near-zero latency.

Despite these technological leaps, practical clinical adoption remains fragmented. Most existing healthcare AI applications either function as isolated text chatbots or demand expensive local GPU hardware (16GB+ VRAM) that cannot be deployed efficiently in lightweight web environments or rural clinics. This creates a compelling need for an integrated system that connects modern cloud inference engines (Groq LPUs, Deepgram API) into a unified, lightweight, and accessible web application.

1.3 Problem Statement

Patients encountering sudden dermatological conditions (e.g., acute rashes, localized erythema, sudden hives, or acne flares) require immediate preliminary guidance to evaluate urgency and alleviate health anxiety. However, several critical issues persist:

- **Consultation Delays:** Long scheduling queues at dermatology clinics delay early triage and treatment.
- **Communication Friction:** Typing detailed medical complaints on mobile keyboards is tedious and error-prone; voice communication is far more intuitive.
- **Visual Dependency:** Dermatological assessment is impossible through text alone; visual examination of lesion borders, elevation, and color is essential.

- **Unreliable Self-Diagnosis:** Patients searching generic search engines often encounter worst-case medical scenarios, leading to panic or harmful self-medication.

The problem addressed by this training project is therefore to design, implement, and validate an automated, multimodal consultation pipeline that accepts natural patient speech and skin imagery/video, performs intelligent visual-linguistic analysis, and responds with spoken, audio-safe clinical guidance under strict ethical guardrails.

1.4 Project Objectives

The core technical and developmental objectives of this training project are summarized in Table 1.1:

No.	Objective	Target / Success Metric
1	Speech-to-Text Integration	Accurately transcribe patient voice into text within 1.5 seconds using Groq Whisper-large-v3.
2	Computer Vision Preprocessing	Implement OpenCV frame extraction for videos and PIL in-memory image downscaling (1024x1024).
3	Multimodal Clinical Reasoning	Integrate Qwen Vision on Groq to evaluate visual lesions alongside transcribed patient complaints.
4	Clinical Prompt Engineering	Enforce strict system prompt guardrails (2-3 sentences, zero markdown symbols, non-diagnostic persona).
5	Neural Speech Synthesis	Convert doctor text advice into natural spoken audio via Deepgram Aura and provide gTTS fallback.

Table 1.1: Project Objectives and Success Metrics

1.5 Scope of the Project

The scope of this project encompasses the design, implementation, and empirical testing of a client-server multimodal application specifically calibrated for dermatological consultation assistance. The technical boundaries include:

- **Voice Ingestion:** Supporting both live microphone audio recording and uploaded digital audio files (.wav, .mp3, .m4a).
- **Visual Media Ingestion:** Supporting static photographic images (.jpg, .png) and short video recordings (.mp4, .mov) with automated frame extraction.
- **Cloud AI Orchestration:** Chaining external high-speed APIs (Groq Cloud and Deepgram) without storing sensitive patient data permanently.
- **Safety & Ethics:** Embedding visible clinical disclaimers across the web interface and enforcing system prompt guidelines to ensure the system is strictly informational.
- **Web GUI Deployment:** Providing an intuitive Gradio Blocks interface featuring modern medical CSS and reactive event triggers.

CHAPTER 2
LITERATURE REVIEW & TECHNOLOGICAL FOUNDATIONS

2.1 Evolution of Clinical Artificial Intelligence

The evolution of Artificial Intelligence in medicine can be categorized into three distinct eras: rule-based expert systems, discriminative deep learning classification, and modern multimodal generative AI. In the first era, systems such as MYCIN employed static production rules and decision trees to evaluate clinical parameters. While groundbreaking, these systems lacked adaptability and were incapable of interpreting unstructured inputs.

In the second era, Convolutional Neural Networks (CNNs), such as ResNet and DenseNet, achieved human-level accuracy in specific image classification tasks, such as distinguishing melanomas from benign nevi. However, these discriminative models acted as 'black boxes'—they produced a statistical classification label without the ability to hold a conversational dialogue, consider patient history, or explain findings in natural language.

The present era is defined by Foundation Multimodal Models that bridge visual understanding with natural language reasoning and acoustic generation. This paradigm allows systems to perceive patient inputs holistically, mimicking the multimodal observation workflow of a human physician.

2.2 Automatic Speech Recognition & Whisper Architecture

Automatic Speech Recognition (ASR) converts acoustic pressure waves into linguistic text tokens. OpenAI’s Whisper model represents a landmark breakthrough in robust speech recognition. Unlike previous acoustic models trained on narrow datasets, Whisper was trained on over 680,000 hours of diverse audio gathered from the internet.

The architecture follows an encoder-decoder Transformer structure. Incoming audio is sampled at 16,000 Hz and processed through a feature extractor that converts the signal into 80-channel log-magnitude Mel-spectrograms using 25ms window sizes with 10ms hop lengths. The encoder consists of stacked Transformer blocks using 1D convolutional layers with GELU

activations to downscale the acoustic features. The decoder predicts text tokens autoregressively, incorporating cross-attention over the encoder's representations.

In this project, Groq's hosted Whisper-large-v3 model is utilized. Operating on specialized Language Processing Units (LPUs), it executes transcription with unprecedented speed (< 1 second for a 15-second speech sample) while exhibiting superior resilience against background noise and clinical jargon.

2.3 Multimodal Large Language Models (Qwen Vision)

Multimodal Large Language Models (MLLMs) integrate vision encoders with large language model backbones to perform visual-linguistic reasoning. Qwen Vision incorporates a Vision Transformer (ViT) that processes visual media as sequences of non-overlapping image patches.

These visual patch embeddings are projected into the input space of the autoregressive language model via cross-attention adapter layers. This allows the model to treat visual patches analogously to text tokens. During consultation, the model evaluates visual dermatological features—such as erythema intensity, lesion elevation, border regularity, and scaling—concurrently with the transcribed patient complaints, enabling holistic preliminary guidance.

2.4 Neural Speech Synthesis (Deepgram Aura)

Speech synthesis has evolved from concatenative methods—which spliced together pre-recorded phoneme units, resulting in unnatural robotic cadences—to modern Neural Vocoders. Neural Text-to-Speech models directly map text character sequences to Mel-spectrogram representations, which are subsequently synthesized into time-domain waveforms using neural diffusion or autoregressive vocoders.

Deepgram Aura represents the state-of-the-art in low-latency conversational speech synthesis. The aura-asteria-en model employed in this application was trained on professional acoustic recordings, delivering empathetic, warm clinical prosody, realistic breathing pauses, and natural speech inflection. Its streaming API architecture enables chunked audio generation, allowing playback to begin before the entire response is synthesized.

2.5 Computer Vision & Frame Extraction with OpenCV

In dermatological assessment, video media offers distinct clinical advantages over static photographs, as patients can showcase lesion elevation, light reflection, and surrounding skin texture across varying angles. OpenCV (Open Source Computer Vision Library) provides optimized C++ and Python bindings for real-time video stream handling.

Through cv2.VideoCapture, video files (.mp4, .mov) are decoded frame-by-frame. Because OpenCV decodes frames into the BGR (Blue-Green-Red) color space by default, mathematical conversion to RGB (Red-Green-Blue) via cv2.cvtColor(frame, cv2.COLOR_BGR2RGB) is required before forwarding to Python Pillow (PIL) and the vision language model.

2.6 Clinical Prompt Engineering Design Principles

Prompt engineering in clinical artificial intelligence serves as a strict deterministic control mechanism. Foundation models, if unconstrained, tend to generate verbose, academic dissertations laden with markdown headings and bullet points. In our voice-driven pipeline, this introduces severe acoustic degradation.

To ensure production quality, five core prompt engineering principles were implemented:

- **Role Conditioning:** Assigning an authoritative yet empathetic clinical persona ('You are a confident, natural doctor specializing in skin care').
- **Conciseness Guardrail:** Restricting response length to strictly two or three sentences, keeping spoken audio under 15 seconds.
- **Acoustic / Spoken Formatting Ban:** Banning special characters, markdown symbols (*, #), and emojis, preventing the speech synthesizer from pronouncing symbol names aloud.
- **Ethical Disclaimer Guardrail:** Instructing the system to provide general informational assistance rather than definitive diagnosis or prescription.

CHAPTER 3
TRAINING WORK UNDERTAKEN & SYSTEM IMPLEMENTATION

3.1 Project Overview

The four-week training program was dedicated to translating theoretical deep learning concepts into an operational, production-ready healthcare application. The work was organized around a structured four-phase developmental schedule, moving systematically from environment architecture and secret isolation to component development, pipeline chaining, and responsive UI engineering.

3.2 System Architecture & Layered Design

The application follows a decoupled, layered architecture to guarantee maintainability, testability, and fault isolation. The high-level architecture comprises five primary tiers:

- **1. Client / Presentation Tier (main.py):** Constructed using Gradio Blocks, providing audio recording inputs, media upload zones, and reactive response cards.
- **2. Audio Transcription Tier (voice_of_patient.py):** Manages audio stream buffering and interfaces with Groq's Whisper-large-v3 transcription endpoint.
- **3. Media Preprocessing Tier (brain_of_the_doctor.py):** Downscales high-resolution images via Pillow and extracts video keyframes via OpenCV, serializing inputs into Base64 strings.
- **4. Multimodal Intelligence Tier (Groq Qwen Vision):** Fuses speech transcripts with visual features to synthesize clinical advice.
- **5. Speech Synthesis Tier (voice_of_the_doctor.py):** Converts clinical text into spoken MP3 audio via Deepgram Aura.

3.3 Technology Stack & Environment

The project relies on a modern Python 3.13 toolchain. Virtual environments were managed using uv, which accelerates dependency resolution and guarantees lockfile reproducibility via uv.lock. Credentials were safeguarded through python-dotenv, ensuring that private API keys

are never hardcoded or committed to version control. The comprehensive technology stack is documented in Table 2.1:

Technology / Tool	Version	Role in Project Architecture
Python	3.13	Primary programming language and runtime environment
Groq SDK	>=1.6.0	Cloud inference interface for Whisper STT and Qwen Vision
Deepgram SDK	>=7.6.0	Neural Text-to-Speech generation via Aura API
OpenCV (cv2)	>=5.0.0.93	Video stream reading, frame decoding, and BGR-to-RGB conversion
Pillow (PIL)	>=10.0.0	Image resizing, thumbnailing, and in-memory JPEG compression
Gradio	>=6.20.0	Web application interface framework with custom CSS
python-dotenv	>=1.2.2	Secure environment variable loading from local .env files

Table 2.1: Comprehensive Technology Stack

3.4 Major System Modules

The repository architecture is organized into distinct, decoupled Python modules as documented in Table 2.2:

File Name	Primary Responsibility	Key Functions / Classes
voice_of_patient.py	Transcribing patient speech to text via Groq Whisper API	transcribe_patient_voice()
brain_of_the_doctor.py	Multimodal reasoning, image encoding, video frame grab	brain_of_the_doctor(), encode_image_for_groq()
voice_of_the_doctor.py	Neural text-to-speech generation via Deepgram Aura	convert_text_to_doctor_audio()
free_text_to_speech.py	Offline zero-cost fallback TTS using Google TTS	text_to_speech_with_gtts()
main.py	Gradio web interface layout, CSS, controller function	process_inputs(), iface.launch()

Table 2.2: Major Project Modules and Source Files

3.5 Detailed Component Implementation

3.5.1 Patient Speech-to-Text (voice_of_patient.py)

The voice transcription module interfaces with Groq's hosted Whisper Large-v3 model. When a patient records their symptoms through the Gradio microphone widget or uploads a prerecorded audio file, the module performs the following operations:

- **Credential & File Validation:** Verifies that GROQ_API_KEY is present in the environment and that the audio file exists at the specified path.
- **Binary Streaming:** Opens the audio file in binary read mode ('rb') and streams the payload to client.audio.transcriptions.create().
- **Text Parsing:** Parses the returned Transcription object and extracts the text property.

3.5.2 Clinical Reasoning & Vision (brain_of_the_doctor.py)

The clinical reasoning core orchestrates multimodal evaluation by combining the transcribed patient speech with the visual media. Key sub-processes include:

- **Image Preprocessing:** Incoming images are resized to a maximum bounding box of 1024x1024 pixels using image.thumbnail((1024, 1024)). The image is converted to standard RGB and compressed to JPEG with quality=75 using an in-memory BytesIO buffer, reducing payload size by over 80%.
- **Base64 Encoding:** The buffer bytes are Base64 encoded and wrapped into a standardized Data URI string: f'{image_data}'.
- **Multimodal API Call:** The Groq client.chat.completions.create endpoint is invoked with model='qwen/qwen3.6-27b' (or 'qwen/qwen3.8-27b'). The payload incorporates the clinical system prompt and user message containing both textual prompt and image URL.
- **Regex Sanitization:** Modern reasoning models often generate internal deliberation tokens (<think>...</think>). To prevent these tokens from being read aloud by the audio synthesizer, regex filtering is applied: re.sub(r'<think>.*?(?:</think>|$)', '', content).strip().

3.5.3 Video Frame Extraction & Preprocessing

To provide maximum flexibility for patients, the application accepts short video clips of affected areas. The function extract_frame_from_video(video_filepath) utilizes OpenCV (cv2):

- **Video Capture Initialization:** Initializes cv2.VideoCapture(video_filepath) to decode the video stream.
- **Keyframe Grab:** Reads the initial valid frame (cap.read()), releases the capture object, and verifies frame integrity.
- **Color Space Conversion:** OpenCV reads video frames in BGR color space. The frame is converted to standard RGB using cv2.cvtColor(frame, cv2.COLOR_BGR2RGB) and instantiated as a PIL Image object via Image.fromarray(frame_rgb).

3.5.4 Neural Speech Synthesis (voice_of_the_doctor.py)

The voice synthesis module translates the doctor's sanitized textual guidance into human-sounding speech using the Deepgram Aura SDK:

- **Voice Model Selection:** Utilizes the aura-asteria-en model, calibrated for an empathetic, professional female clinician tone.
- **Payload Protection:** Deepgram limits API text requests to 2,000 characters. The module defensively slices incoming text via text = text[:1999] to prevent buffer overflow exceptions.
- **Streaming Chunk Buffer:** The audio generator iterates over binary chunks from deepgram.speak.v1.audio.generate and writes them directly to doctor_response.mp3.

3.5.5 Fallback Speech Synthesis (free_text_to_speech.py)

As a contingency against cloud service rate limits or credential exhaustion, an offline fallback script was implemented using gTTS (Google Text-to-Speech). It generates an MP3 file locally and uses cross-platform sub-processes (such as PowerShell SoundPlayer on Windows, afplay on macOS, or aplay on Linux) to trigger automatic playback.

3.5.6 Web Interface & Master Controller (main.py)

The graphical user interface is constructed using Gradio Blocks. It employs a two-column responsive desktop layout:

- **Left Column (Patient Input):** Contains an interactive Audio widget (microphone or upload), media upload boxes for skin images and videos, and a high-visibility 'Analyze Concern' primary button.
- **Right Column (Doctor Response):** Displays the transcribed patient speech, the doctor's guidance textbox, and an auto-playing Audio output widget.
- **Master Controller:** The function process_inputs(audio_filepath, image_filepath, video_filepath) orchestrates sequential execution: validating presence of inputs, calling voice_of_patient, forwarding text to brain_of_the_doctor, and feeding output to voice_of_the_doctor.

3.6 Implementation Methodology

The project implementation followed an Agile, iterative engineering methodology across the 4 weeks:

- **Week 1 (Architecture & Setup):** Establishment of repository structure, dependency configuration with pyproject.toml, and verification of Groq and Deepgram API connectivity.
- **Week 2 (Audio Pipeline):** Development of voice_of_patient.py, testing Whisper transcription across multiple accents, and measuring latency.
- **Week 3 (Vision & Prompting):** Integration of OpenCV frame extraction, PIL thumbnailing, Qwen Vision multimodal calls, and regex sanitization.
- **Week 4 (Audio Synthesis, UI & Deployment):** Deepgram Aura SDK integration, buffer management, Gradio interface construction with custom medical CSS, and end-to-end testing.

3.7 Validation, Error Handling & System Reliability

Defensive programming practices were embedded across every layer of the application to ensure graceful failure recovery:

- **Input Presence Validation:** If a patient clicks 'Analyze Concern' without providing audio or visual media, Gradio raises gr.Error('Please record or upload your voice description first.') rather than crashing the backend.
- **Missing API Key Detection:** Modules inspect os.environ.get('GROQ_API_KEY') and os.environ.get('DEEPGRAM_API_KEY') at startup, raising immediate informative exceptions if credentials are absent.
- **Video Frame Verification:** If a corrupted video file is uploaded where cap.read() returns success=False, a descriptive ValueError is thrown preventing null pointer exceptions downstream.

CHAPTER 4
RESULTS, TESTING & DISCUSSION

4.1 Testing Approach & Methodology

The testing strategy was designed to rigorously evaluate input-output validity, end-to-end response latency, transcription fidelity across diverse accents, and visual diagnostic consistency. Testing was conducted using the local Test data/ repository comprising real dermatological condition images (rashes, acne, eczema, and localized irritation) paired with varied audio descriptions.

4.2 Representative Input-Output Results

Representative consultation scenarios illustrating the end-to-end performance of the application are documented in Table 3.1:

Case ID	Patient Voice Input	Visual Media	Doctor Guidance Output
Case 1 (Eczema)	"I have had this red, itchy patch on my forearm for three days, and it feels dry."	Forearm skin photograph with localized erythematous scaling	"This appears consistent with mild eczema or localized dermatitis. Keep the area moisturized with a fragrance-free cream and consult a dermatologist if redness spreads."
Case 2 (Acne)	"I have painful breakouts on my chin that started appearing last week."	Close-up photograph showing inflammatory papules on chin	"The visual indicates inflammatory acne vulgaris. Cleanse gently twice daily with a mild wash, avoid picking the lesions, and seek medical advice for targeted topical care."
Case 3 (Video)	"My ankle has been itching severely after hiking yesterday."	Short 4-second video clip showing ankle irritation	"Based on the video frame, the mild swelling and erythema suggest a possible contact dermatitis or insect reaction. Apply a cold compress and visit a clinic if swelling worsens."

Table 3.1: Representative Input-Output Test Cases

4.3 Performance & Latency Benchmarks

Latency benchmarking was conducted across 15 trials under standard broadband internet connectivity (50 Mbps). The timing breakdown across individual pipeline stages is summarized in Table 3.2:

Pipeline Stage	Average Latency (s)	Observations / Throughput
Speech-to-Text (Whisper)	0.85 s	Groq LPU acceleration processes 15s audio in < 1s
Video Frame Extraction	0.12 s	OpenCV decodes first valid keyframe with minimal overhead
Vision Reasoning (Qwen)	1.20 s	Multimodal token generation bounded to ~60 tokens
Speech Synthesis (Aura)	0.75 s	Streaming chunk generation completes before playback starts

Table 3.2: End-to-End Latency and Performance Benchmarks

The total end-to-end round trip time averages approximately 2.92 seconds, which is well within acceptable thresholds for conversational medical triage interfaces.

4.4 Prompt Engineering & Sanitization Observations

Several critical observations emerged regarding prompt structure and LLM behavior:

- **Formatting Stripping:** Without explicit prompt constraints banning markdown, the LLM frequently outputted bold text ('**Diagnosis:**') or bullet points. When fed into Deepgram, the synthesizer read out 'asterisk asterisk Diagnosis asterisk asterisk', ruining voice naturalness. Banning markdown completely solved this.
- **Reasoning Token Excision:** Reasoning models occasionally leaked internal thinking blocks ('<think> The user has a red rash... </think>'). Implementing regex sanitization re.sub(r'<think>.*?(?:</think>|$)', '', content) ensured that only final guidance reached the patient.
- **Sentence Capping:** Restricting responses to 2–3 sentences prevented verbose textbook dissertations, keeping the audio playback under 15 seconds.

4.5 Technical Challenges & Engineering Solutions

From an engineering perspective, several key architectural insights were verified:

- **In-Memory Image Buffering:** Using PIL with io.BytesIO completely eliminated disk I/O bottlenecks. Temporary image files were never written to disk, enhancing performance and ensuring patient privacy.
- **LPU Acceleration:** Groq's Language Processing Units offered a 5x speedup compared to standard cloud GPU instances, critical for preventing user drop-off during consultations.
- **Decoupled Fallback:** Maintaining free_text_to_speech.py with gTTS proved essential during API maintenance windows.

4.6 Medical AI Ethics, Safety & Limitations

While highly functional, the current system exhibits certain boundaries that must be recognized:

- **Single Video Frame:** The video extraction module grabs only the first clear frame. Conditions that require multi-angle 3D inspection may not be fully evaluated.
- **Lighting & Camera Quality:** Blurry, under-exposed, or low-resolution camera images degrade vision model confidence.
- **Internet Dependency:** The system requires an active broadband connection to communicate with Groq and Deepgram cloud APIs.
- **Informational Nature:** The system cannot provide definitive prescriptions or replace biopsy-confirmed clinical diagnosis.

CHAPTER 5
CONCLUSION & FUTURE SCOPE

5.1 Conclusion

The four-week training program (TR-102) at DigiMantra Labs provided an intensive, hands-on opportunity to master modern applied Artificial Intelligence, speech technologies, computer vision, and prompt engineering. By anchoring the training around a tangible, socially impactful healthcare problem, theoretical machine learning concepts were successfully transformed into a robust, working application.

The resulting minor project, AI Skin Specialist, demonstrates the immense power of multimodal AI orchestration. By chaining OpenAI Whisper for voice transcription, Groq Qwen Vision for clinical reasoning, OpenCV for video keyframe extraction, and Deepgram Aura for neural voice synthesis, the application establishes a seamless, low-latency (< 3 seconds) consultation flow. Crucially, the implementation balances technological innovation with strict ethical guardrails, enforcing defensive input validation, prompt constraints, and medical disclaimers.

In summary, the project demonstrates that state-of-the-art Generative AI can be harnessed to create intuitive, accessible, and responsive healthcare assistance tools that bridge communication barriers and assist patients prior to formal clinical evaluation.

5.2 Future Scope

The current application establishes a solid baseline for multimodal clinical triage. Several promising enhancements are identified for future development:

- **Multilingual Speech Support:** Integrating multilingual ASR and TTS to allow rural patients to consult in regional languages such as Hindi, Punjabi, Bengali, or Spanish.
- **Multi-Frame Temporal Video Analysis:** Enhancing the OpenCV pipeline to sample frames across the entire duration of a video clip, analyzing lesion progression from multiple lighting angles.

- **Consultation PDF Summary Generation:** Developing an automated reporting module that compiles the patient transcript, timestamped skin photograph, and doctor advice into a downloadable PDF for the patient's real doctor.
- **Offline Edge Deployment:** Quantizing lightweight foundation models (e.g., Whisper-Tiny and MobileNet) to run locally on low-power edge devices and smartphones without requiring internet connectivity.
- **Multi-Turn Conversational Memory:** Extending the Gradio controller with conversation history so patients can ask follow-up questions regarding home care and symptom progression.

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APPENDIX A
SYSTEM PROMPT & ARCHITECTURAL SPECIFICATIONS

This appendix contains the production system prompt templates and core controller logic utilized in the AI Skin Specialist application.

A.1 Clinical System Prompt Specification

```
ROLE & PERSONA DEFINITION:
You are a confident, natural doctor specializing in skin care. Speak with
the reassurance, clarity, and authority of a real doctor.

LENGTH CONSTRAINTS:
Limit your entire response to two or three sentences maximum.

FORMATTING & ACOUSTIC RULES:
Do not use any special characters, symbols, asterisks, or markdown
formatting in your response because it will be converted directly to
audio.

SAFETY GUARDRAIL (SYSTEM LAYER):
You are a careful skin care assistant. Give general information, not a
diagnosis.
```

A.2 Video Keyframe Extraction Code Structure

```
def extract_frame_from_video(video_filepath):
    import cv2
    cap = cv2.VideoCapture(video_filepath)
    success, frame = cap.read()
    cap.release()
    if success:
        frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
        return Image.fromarray(frame_rgb)
    return None
```

A.3 Regex Sanitization Pattern

```
# Excising internal reasoning deliberation blocks from model output
content = re.sub(r'<think>.*?(?:</think>|$)', '', content,
flags=re.DOTALL).strip()
```